The Quiet Half of the Blast Radius: LLMs Catch the Blackhole but Miss the Silent Reroute in Network Configuration Changes

Technical Report v1

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June 2026

Abstract

Large language models are beginning to write network configuration changes, and the safety question has so far been posed as a binary: *is the change safe or not?* We argue the operational question is graded and typed: *exactly which BGP sessions, prefixes, paths, and forwarding behaviors does this change damage?* We introduce *blast-radius prediction*: given device configurations and a proposed diff, a model must enumerate the typed set of damaged elements across nine scored damage dimensions—from BGP sessions, prefixes, and reachability to waypoint and redundancy loss, forwarding anomalies, resource exhaustion, control-plane withdrawals, and intent violations. Ground truth is computed by the Batfish network verifier on a digest-pinned, determinism-gated snapshot pair, and predictions are scored by deterministic, semantics-aware set comparison—no LLM judge. We evaluate eight models from five vendors—six proprietary and two open-weight reasoning models—on NetBrace, a benchmark of 11 incident-grounded scenarios and 48 generator-produced instances with randomized identifiers: 1,070 graded samples (N=5 per instance). The measured capability split is sharp and runs along an operationally meaningful line. Models read the *loud* half of the blast radius well: five of the eight identify both grounded blackholes—the explicit leak-discard and the second-order partial-ECMP one—in every sample (10/10), a sixth (Sonnet 4.6) at F1 0.97, and they read first-order prefix loss at F1 0.67–0.83. The *quiet* half—damage that removes no nameable object and leaves reachability intact—*separates the fleet*: a leak-induced silent reroute that pushes a victim's traffic off its mandated waypoint, while every route still resolves and every session stays up, is missed outright by most models—scoring zero on both the waypoint and intent dimensions on a majority of samples, including the loud-half leaders Opus 4.8 (21 of 30) and Gemini 3.5 Flash (all 30). Only two clear it: GPT-5.5 (misses both on 1 of 30; waypoint F1 0.64) and the open-weight DeepSeek v4-pro (10 of 30; 0.56), statistically tied at the top and well above the rest. An open-weight reasoner standing among the two leaders—ahead of every Anthropic and Google model evaluated—indicates the gap is a reasoning trait, not a closed-model or grader-circularity artifact; the differentiation is itself grader-dependent (under naive exact-string matching every model misses all 30, the same granularity artifact we report for forwarding anomalies). The damage models *do* catch shares a signature: it removes or blackholes something they can name—a withdrawn route, a vanished management prefix, a blackholed destination, which the stronger models catch at F1 1.00 where each is the ground truth (the two cheapest lower, 0.00–0.80, with GPT-5.4-mini missing the lockout entirely). No model ever raises a false session-loss alarm (0/1,070). The most expensive real outages of the last decade—leak-induced silent reroutes among them—live in the quiet half. We additionally show that naive exact-string set grading would have reported a uniform 0.00 on forwarding anomalies for every model—a granularity artifact, not a capability

fact—and release both graders with a frozen semantic-matching specification. The benchmark, oracle pipeline, and run artifacts will be released.

1 Introduction

Network operators are rapidly adopting LLM-based assistants that propose, review, and increasingly *apply* configuration changes. The emerging evaluation literature measures whether these systems can *fix* things: configuration synthesis from intent [6], end-to-end repair of injected misconfigurations [2, 1], multi-turn configuration tasks [4], and incident troubleshooting [5]. Where safety appears, it is binary or aggregate: NetArena scores rule-adherence of agent actions in an emulator [3]; the agentic-repair study of Asadli et al. [1] reports a regression score—the fraction of the checked specification set that an *applied* repair newly breaks—computed *by the verifier* after the fact.

Production change review asks a different question. Before a change ships, a reviewer is expected to articulate its *blast radius*: which BGP sessions will drop, which prefixes will disappear from which routing tables, which flows will lose their path, which traffic will silently blackhole, and which device will exhaust its FIB. The damage is *typed* and *itemized*, and the most expensive failures in operational history are *quiet*: the configuration change looks innocuous, every BGP session stays established, often every destination even stays reachable—and the damage emerges only from the interaction of routing policy with topology. A leaked more-specific route attracts traffic onto a path that cannot deliver it—congesting it, as in the 2019 route-leak incidents, or discarding it outright, as we model here for determinism. A "cleanup" filter's implicit deny silently withdraws a customer prefix while the session stays up. A LOCAL_PREF adjustment quietly steers traffic around the inspection node it is required to transit.

This paper asks whether LLMs can see that damage *before* a change is applied, from the same evidence a human reviewer has: the device configurations and the diff. We make three contributions.

1. Task: typed blast-radius prediction. Given pre-change configurations and a proposed diff, the model must output the affected *set*—typed elements across nine scored damage dimensions (§2)—and is scored with per-dimension precision, recall, and F1 against verifier-computed ground truth. Correctly predicting *no* damage is a first-class outcome (several instances have empty ground truth by construction), and the cliché failure mode—hallucinating a BGP session loss—is tracked as an explicit false-session-alarm metric.

2. Benchmark and grading methodology: NetBrace. A two-split question base (§3): 11 hand-authored scenarios grounded in documented outage classes (route-reflector filter withdrawals, more-specific leaks, Clos summarization blackholes, FIB-overflow de-aggregation, management-plane lockout), plus 48 instances stamped by four parameterized generators that re-randomize ASNs, addressing, and names per seed as a contamination control. Ground truth is computed by Batfish [11] on a digest-pinned engine; every instance passes a $\geq 10\times$ byte-identical determinism gate before any paid model call. Grading is deterministic set arithmetic with *semantic canonicalization* (§2)—no LLM judge anywhere in the loop [14]. We show this canonicalization is not optional: naive exact-string matching against probe-granularity ground truth fabricates a uniform 0.00 on the forwarding-anomaly dimension for every model (§2, Appendix A)—a measurement pitfall we expect generalizes to other verifier-graded benchmarks.

3. Measurement: the quiet half of the blast radius is the blind spot. Across eight models from five vendors (GPT-5.5, GPT-5.4-mini, Claude Opus 4.8, Sonnet 4.6, Haiku 4.5, Gemini 3.5 Flash, and the open-weight DeepSeek v4-pro and Kimi k2.6; N=5 samples per instance; 1,070 graded samples), capability splits along the loud/quiet line, not the first-order/second-order line we expected. Loud damage is read well: five of the eight models—including both open-weight reasoners—identify both grounded blackholes (including the second-order, partial-ECMP one) on every sample, a sixth (Sonnet 4.6) at D7 F1 0.97, first-order prefix loss is read at F1 0.67–0.83, and damage that *removes* a nameable object—a withdrawn route, a vanished management prefix, a dropped ECMP path—is caught at F1 1.00 (stronger models; 0.00–0.80 for the cheapest, mini missing the lockout) where it is the ground truth. Quiet damage—where reachability survives and nothing the diff directly changes looks broken—*separates the fleet*: traffic rerouted past a mandated waypoint by a leak *elsewhere*, with every route still resolving, is missed by most models (Opus 4.8 scores zero on both the waypoint and intent dimensions on 21/30 and Gemini 3.5 Flash on all 30—both despite F1 1.00 on the loud half) and cleared only by two models (GPT-5.5 misses 1/30, the open-weight DeepSeek v4-pro 10/30)—as is the traffic-pinning intent a surviving covering aggregate hides even as the pinned specifics are dropped (§5). The split is not first- vs. second-order (the Clos blackhole is second-order and caught); it is whether the damage leaves a trace. The errors are one-sided: the false-session-alarm rate is 0.00 across all samples. Model-free floors (empty, naive, syntactic-diff, and a per-class memorizer) score $\leq$0.02 grounded and $\leq$0.08 templated, confirming the benchmark measures reasoning headroom rather than template recall; and reading the silent reroute is shared by one proprietary and one open-weight model (GPT-5.5 and DeepSeek), the open one ahead of every Anthropic and Google model, ruling out a closed-model or Claude-family grader artifact on exactly the capability the benchmark most sharply discriminates.

We argue this split—competent at damage that announces itself in the routing table, blind to damage that hides behind preserved reachability—motivates a concrete *candidate graduation test* on the autonomy ladder for network agents (§6): before a system self-approves changes, it should first demonstrate it can enumerate the blast radius of its own proposals, *including the quiet half*—a bar no evaluated model currently meets.

2 Blast-Radius Prediction

Setting. A task instance is a triple: a set of device configurations C (one file per router, FRR dialect in this release), a proposed change expressed as a unified diff Δ over one or more devices, and a natural-language statement of operator intent. The model receives all three plus an output contract, in a single turn, with no tools; it must return a JSON object enumerating the damaged elements. The no-tools condition is deliberate: it isolates the model's *own* causal model of the network—the capability a pre-deployment reviewer exercises— from its ability to drive external tools, and it reflects the common deployment where an assistant reviews changes without a verifier in the loop. (A tools-augmented contrast is explicitly future work; §6.)

Damage dimensions. Damage is typed. Each element type defines a canonical string key (e.g., `bgp|vrf=default|node=r1|peer_ip=10.0.0.4`), making ground truth and predictions comparable as sets. Ten dimensions are declared; D6 is deferred in this release, so **nine are scored**, and every result in this report is over those nine:

- **D1 sessions**: BGP sessions established before, not after.
- **D2 prefixes**: watched prefixes present in a device's main RIB before, absent after.

- **D3 reachability**: flows delivered before, undeliverable after.
- **D4 waypoints**: flows that must transit a waypoint (e.g., an inspection node) and no longer do.
- **D5 redundancy**: (node, prefix) pairs whose installed equal-cost path count drops.
- **D6 packet filters**: permit↔deny transitions (*deferred* in this release: Batfish does not model FRR access-lists as interface filters; a vendor dialect is in progress).
- **D7 forwarding anomalies**: flows newly blackholed, null-routed, or looping while a route for them persists.
- **D8 scale**: post-change installed-route counts crossing a declared per-device resource budget (FIB/RIB).

Two auxiliary control-plane surfaces are scored alongside: **D9** BGP-RIB best-path withdrawals (control-plane loss masked from the FIB by, e.g., a static backup) and **D10** violations of declared intent invariants (reachability, waypoint, isolation, redundancy-floor, and traffic-pinning assertions evaluated oracle-side and never shown to the model).

Oracle. Ground truth is computed by Batfish [11] on a *snapshot pair*: the pre-change configurations and the post-change configurations produced by strict application of Δ. Sessions, RIBs, BGP-RIBs, route counts, and flow traces are queried in both worlds and diffed per dimension. Three properties make the oracle trustworthy as an answer key: (i) the engine is pinned by container digest, so ground truth is tied to exact bits; (ii) every instance passes a determinism gate—the full computation is repeated $\geq 10\times$ on fresh networks and must produce byte-identical key sets; (iii) scenario classes were spot-validated against live FRR containers (sessions, RIBs, traceroutes) before being trusted.

Grading. For each dimension we compute precision, recall, and F1 over element sets, with the empty-set convention: if truth and prediction are both empty the dimension scores 1.0 (correctly predicting *no* damage); a non-empty prediction against empty truth scores 0 (false alarm); an empty prediction against non-empty truth scores 0 (miss). The instance score is macro-F1 over *active* dimensions (non-empty in truth or prediction), so models cannot farm score by emitting empty lists for irrelevant dimensions; per-dimension means follow the same active convention.

Element matching is *semantics-aware*, per a specification frozen before any per-model number was computed (released with the artifacts). The oracle expresses flow-scoped truth (reachability, waypoints, forwarding anomalies, some intents) as probe *host routes* (/32s), but the output contract asks models for destination *prefixes*—the granularity the configurations actually expose. We therefore canonicalize each probe to its most-specific config-visible covering prefix and credit a prediction whose prefix lies in the band [probe, canonical cover]; all non-prefix fields (node, source, waypoint, VRF) still match exactly, and matching is deterministic one-to-one bipartite. The band is two-sided and not gameable: a prediction broader than the canonical cover earns nothing (no credit for `0.0.0.0/0`; an over-broad answer claims demonstrably-undamaged traffic is damaged and is charged as a false positive), and a narrower prediction earns credit only if it covers the probed host—which requires correctly reasoning about *which* traffic the change harms. For forwarding anomalies, `blackhole` and `null_routed` form one equivalence class (the contract's own gloss conflates them); `loop` stays distinct. The canonicalization is provably the identity on dimensions whose truth is already config-visible (D1, D2, D8, D9), and it leaves every model-free floor at

chance: the largest floor movement under it is the syntactic-diff floor's 0.007→0.023 (templated), with all floors remaining ≤0.02 grounded and ≤0.08 templated (§3.3).

Two flags are tracked independently: *false session alarm* (predicting D1 loss when truth has none—the cliché reflex) and *invalid output* (no parseable JSON; reported separately so formatting failure is never conflated with reasoning failure). Parsing is tolerant (last JSON object in the response; missing keys are empty lists; malformed elements are dropped).

Why naive set grading fails here. Exact-string matching of prediction keys against probe-granularity truth keys is *unwinnable by construction*: a model that names the correct node, the correct anomaly kind, and the exact prefix the configuration exposes (e.g., the leaked /25) still scores 0 against a truth key carrying an undisclosed probe /32 inside that prefix. Under exact-string grading, the forwarding-anomaly F1 (active convention) is exactly 0.00 for every model—no forwarding-anomaly-truth sample earns any credit, even when the model names the leaked /25 against the undisclosed /32 probe—while the raw outputs show frontier models naming the blackholing node and the affected prefix on every such sample. We report the exact-string numbers in Appendix A as a conservative lower bound and a cautionary result for verifier-graded benchmark design: when an oracle's ground truth and a model's output contract live at different granularities, string-set scoring measures the mismatch, not the model.

3 The NetBrace Benchmark

3.1 Grounded split: 11 incident-anchored scenarios

Each hand-authored scenario instantiates a documented outage class on a small (4–6 node) topology whose configurations are verified against live FRR and whose ground truth passes the determinism gate. Table 1 summarizes them. Nine of eleven are *traps*: the change looks routine, every BGP session stays established (D1 truth is empty), and the real damage lives in D2–D10; the remaining two (*waypoint4*, *ecmp4*) are static-routed sanity anchors with no BGP at all, so their empty D1 is structural rather than adversarial. Examples: an iBGP route-map "cleanup" on a route reflector whose implicit deny silently withdraws a customer prefix (sessions: all established); a leaked more-specific /25 that attracts a victim /24's traffic into a discard (neither session nor prefix counts change); a Clos spine summarization that blackholes exactly one spine's share of ECMP traffic (the blackhole arises from a fixed wiring asymmetry that the summary masks—an instance of the summarization hazard documented in RFC 7938 [12]; the dynamic link-failure variant is future work); a de-aggregation event that overflows an edge FIB budget while reachability is fully preserved; and a management-prefix omission that locks the NOC out of every device (the scenario models in-band management reachability, the common and dangerous case).

3.2 Templated split: 48 generator instances

Four parameterized generators convert verified mechanisms into structurally distinct instances: *silent withdrawal* (fan-out $k \in \{2, 3, 4\}$; one-vs-all peers filtered; in/outbound; plain/static/aggregate prefix backing; implicit-deny vs explicit-deny vs off-by-one prefix-list boundaries), *more-specific leak* (blackhole vs *reroute* variants—in the latter, reachability survives while traffic silently shifts; leaked-scope variants), *attribute path-shift* (LOCAL_PREF and prepend knobs, including no-op variants whose ground truth is *empty* by construction—calibration traps), and *de-aggregation overflow* (clearly-over, marginally-over, and under-budget variants). Every instance re-randomizes ASNs (random private 4-byte), addressing, node names, and the customer prefix per seed, so instances

Scenario	Mechanism class	Damage (truth)	Trap
ipc3	RR cleanup-filter implicit deny	D2, D3 (D1 = ∅)	✓
ipc3_static	same + static backup masks FIB	D9 only	✓
ipc_fanout	filter on whole peer group	D2×3, D3×3	✓
ipc_aggregate	/25s dropped, /24 aggregate survives	D2, D10	✓
leak_ms	more-specific leak → LPM blackhole	D3, D7	✓
clos_summary	spine summarization, partial ECMP blackhole	D2, D7	✓
scale_deagg	de-aggregation overflows FIB budget	D2, D8	✓
mgmt_lockout	mgmt prefixes omitted → NOC lockout	D3, D10	✓
lp_waypoint	LOCAL_PREF shift bypasses inspection	D4	✓
waypoint4	route shift bypasses waypoint (static lab)	D4	
ecmp4	one of two equal-cost paths removed (static lab)	D5	

Table 1: The grounded split. "Trap": the naive answer (a BGP session drops) is wrong; D1 ground truth is empty while damage occurs elsewhere. Every row's damage signature is verified against the committed oracle output. *waypoint4/ecmp4* are BGP-free static-routed anchors.

cannot be answered by recall of public configurations; topology identifiers shown to the model are content-free hashes. The split serves as the statistical backbone (real N per dimension) and the contamination control: a per-class memorizer floor (§3.3) directly tests whether template recognition suffices.

3.3 Model-free floors

Every sweep computes six deterministic baselines on the same instances: *empty* (predict nothing), *naive-session-loss* (predict exactly the cliché: one session drops at the changed device), *naive-count* (a counting heuristic over watched prefixes), *syntactic-diff* (regex over the diff text, zero routing knowledge), *modal-by-class* (replay the per-class median ground truth—a perfect memorizer of the generator's templates), and *oracle-ceiling* (echo the truth; a pipeline sanity check that must score exactly 1.0). Each floor operationalizes a specific skeptical hypothesis about what the benchmark "really" measures. All floors are evaluated under both graders; the semantic canonicalization leaves every floor at chance (§5).

4 Experimental Setup

Models. Eight models from five vendors, pinned IDs as served in June 2026: GPT-5.5 and GPT-5.4-mini (OpenAI), Claude Opus 4.8, Sonnet 4.6, and Haiku 4.5 (Anthropic), Gemini 3.5 Flash (Google), and two open-weight reasoning models—DeepSeek v4-pro (DeepSeek) and Kimi k2.6 (Moonshot AI), served through their hosted OpenAI-compatible APIs. GPT-5.5 and GPT-5.4-mini run with reasoning enabled and accept no sampling parameters; Opus 4.8 has removed them; all are sent no temperature except Haiku 4.5 and Sonnet 4.6 (temperature 0.7) and Gemini 3.5 Flash (API default). DeepSeek and Kimi are reasoning models whose hidden chain-of-thought is streamed and discarded; the open-weight pair is run with streaming and a generous output-token budget (up to 128k, Moonshot's documented reasoning-benchmark recommendation) so a long chain-of-thought is never truncated before the final answer—which is short in practice (under 1k tokens)—and both returned zero invalid grounded samples. Gemini 3.5 Flash is Google's fast tier (the 3.5 Pro tier was not generally available on the API at sweep time); its scores land within noise of Opus 4.8 and GPT-5.5, so we group it with the frontier set. Single-shot prompting under one

frozen prompt contract (frozen, with a leak-lint pass, before any paid call; the contract requires one JSON object with one key per dimension).

Protocol. N=5 samples per (instance, model). All eight models ran the grounded split ($8 \times 11 \times 5 = 440$ samples); the templated split was run on Haiku 4.5 and GPT-5.4-mini ($2 \times 48 \times 5 = 480$ samples) under the study's fixed per-provider cost cap—the full-fleet templated run is gated on the in-progress human-expert validation study. The six-instance silent-reroute class (F3) was run on *all eight models*: Haiku and mini cover it inside the templated split, and the other six (Opus 4.8, Sonnet 4.6, GPT-5.5, Gemini 3.5 Flash, DeepSeek v4-pro, Kimi k2.6) ran it as an extension on the identical six reroute instances ($5 \times 6 \times 5 = 150$ samples complete as of this draft; the Kimi reroute run is finishing), for **1,070 graded samples** to date. Both the prompt contract and the floor definitions were frozen before any paid call. The semantic-grader matching rules (§2) were motivated by the exact-string results and a raw-output audit (which showed models naming the node, kind, and config-visible prefix that the string grader rejected), were specified from the contract/oracle granularity gap rather than tuned against model scores, and were frozen before any per-model number was computed under them; the floor-invariance and config-visible-identity properties (F5) hold regardless of that timeline. Aggregation is two-stage: per-instance mean over samples, then mean ± population std across instances (with SEM = std/$\sqrt{n}$ used for significance)—never pooling samples as i.i.d. Per-dimension cells backed by a single instance are reported as point estimates, never as mean±std. We fix N=5 rather than higher: a variance decomposition shows the grounded macro-F1 standard error is dominated by between-instance spread, not within-instance sample noise (within-instance std ≈0.02–0.03 for frontier models versus 0.06–0.17 across instances), so it changes by <0.001 for any $N > 5$; statistical power is therefore invested in instance count, not samples. Headline damage-detection macros exclude empty-truth calibration instances (an empty predictor passes those for free); the grounded split has no such instances (all 11 carry damage, so its damage-macro equals its full macro), and the calibration traps live in the templated split, scored and reported separately via the false-alarm channel. Total model spend for the study was approximately $45, including the Sonnet grounded re-run, the two open-weight models (grounded and reroute), and the cross-vendor silent-reroute extension, all at the higher token cap (Disclosure ii; §5); per-call costs are logged in every run manifest. The oracle container digest is pinned (`09817554db90…`) and recorded with every run manifest.

Disclosures. (i) Gemini's API returned transient 503 errors on a fraction of calls; its clean N=5 is pooled from three post-freeze runs under the identical frozen prompt and the same pinned model ID (draws are i.i.d. samples of one condition; only Gemini is pooled, and only for provider outages; partial and pooled estimates agree to within 0.01 macro-F1). (ii) The initial sweep capped Anthropic-path output at 4,000 tokens; Sonnet 4.6—the most verbose reasoner in the fleet—hard-truncated on 6/55 grounded samples (all emitting exactly 4,000 tokens, with the correct analysis, including the *clos_summary* blackhole, visible in the cut-off prose). We re-ran Sonnet's grounded split at a 12,000-token cap (parity with the other providers' budgets) under the identical frozen prompt; the re-run yielded zero invalid samples and is the source of Sonnet's reported grounded numbers. The oracle ground truth is byte-identical between the two runs (confirmed by the determinism gate), so only Sonnet's predictions differ; no other model approached the cap. (iii) One registry scenario (a static-route blackhole) was excluded from the paid sweep by a pre-registration audit that judged its topology contrived; it is an explicitly induced discard of the kind every frontier model detects reliably, so including it would have *raised* model scores on D7. (iv) Kimi k2.6's first grounded run hit a streaming/token-budget wedge that left 19 of 38 samples with empty output; we

Model	macro-F1 (damage)	blackhole (D7†)	withdrawal (D9‡)	lockout (D10‡)	false sess. alarm	invalid
Claude Opus 4.8	0.51±0.17	**1.00**	**1.00**	**1.00**	0.00	0.00
Kimi k2.6 *(ow)*	0.49±0.12	**1.00**	**1.00**	0.80	0.00	0.00
Gemini 3.5 Flash	0.49±0.18	**1.00**	**1.00**	0.20	0.00	0.00
GPT-5.5	0.48±0.17	**1.00**	**1.00**	**1.00**	0.00	0.00
DeepSeek v4-pro *(ow)*	0.44±0.08	**1.00**	**1.00**	0.80	0.00	0.00
Claude Sonnet 4.6	0.42±0.06	**0.97**	**1.00**	**1.00**	0.00	0.00
Claude Haiku 4.5	0.39±0.10	0.57	0.80	0.76	0.00	0.00
GPT-5.4-mini	0.27±0.21	0.17	0.60	0.00	0.00	0.00
floors (best of 5)	≤0.02	0.00	0.00	0.00	1.00*	—

Table 2: Grounded split (11 scenarios, N=5, semantic grader); eight models across five vendors, *(ow)* = open-weight. All three columns are *F1 on the instance(s) where the dimension is the ground truth* (recall is at least as high)—the loud half, which models read well. †Mean D7 F1 over the two anomaly-truth instances (*leak_ms*, *clos_summary*); six of eight—including both open-weight models—read both blackholes at ≥0.97, the two cheapest miss the implicit Clos one. ‡F1 on the single grounded instance where the dimension is the truth: the masked control-plane withdrawal (*ipc3_static*, D9) and the management-plane lockout (*mgmt_lockout*, D10). The *quiet* half (the silent reroute, missed) is templated and reported in §5 (F3); the low active-convention D7/D9/D10 *dimensional* means (Figure 1, F3) are precision-dragged by over-prediction, not recall failures (F3). *The naive-session-loss strawman trips the false-session-alarm metric at 1.00, demonstrating the metric detects the reflex; every real model passes it.

re-ran its grounded split with streaming at the open-weight token budget, yielding the zero-invalid set we report. Gemini's reroute set is likewise a clean re-run (an earlier partial had two API-error invalids); both re-runs use the identical frozen prompt and pinned model ID, and Gemini scores 0.00 on the reroute either way.

5 Results

F1: The benchmark discriminates. The eight models span 0.27–0.51 damage-detection macro-F1 on the grounded split: the four leaders (Opus 4.8, Kimi k2.6, Gemini 3.5 Flash, GPT-5.5) cluster at 0.48–0.51, the open-weight pair brackets the mid-tier (Kimi 0.49 tying the top of the fleet, DeepSeek v4-pro 0.44 above both Sonnet 4.6 and Haiku 4.5), and GPT-5.4-mini trails at 0.27; all model-free floors score ≤0.02 on that split (Table 2). On the templated split's damage instances, Haiku 4.5 and GPT-5.4-mini score 0.30±0.18 and 0.21±0.20 while the modal-by-class memorizer scores **0.00** and the best other floor 0.075 (Figure 3): the task is neither impossible nor saturated, and performance is not explained by template recall.

F2: Models do not cry wolf on sessions. The false-session-alarm rate is 0.00 for every model on every split it was evaluated on—all 1,070 samples. The widely-assumed failure mode—hallucinating session losses on any BGP-adjacent change—did not occur once, while the metric itself is demonstrably sensitive (the strawman floor trips it at 1.00). Models do, however, sometimes *double-type* real damage: on silent-withdrawal instances, models often report the dropped traffic both as reachability loss (correct—the frontier scores 1.0 there on the very same samples) and additionally as a forwarding anomaly—a typing the contract's own gloss ("blackhole = no route")

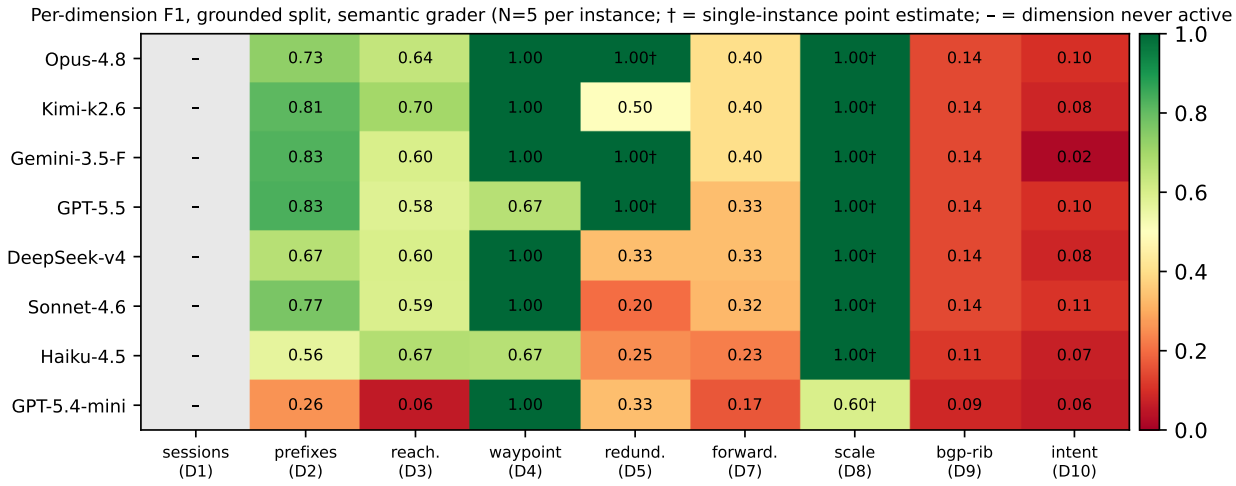


Figure 1: Per-dimension F1 on the grounded split (semantic grader), as *active-convention means*—precision and recall combined over every instance where the dimension is active in truth or prediction. The consequence-dimension cells (D7, D9, D10) are depressed by taxonomy-boundary over-prediction, *not* by missed detection: F1 where each is the ground truth is high (Table 2, F3). The figure is shown to expose that precision gap, not as a detection score.

invites but the oracle reserves for discarded-while-routed flows. We charge these as false positives—and the same taxonomy-boundary over-prediction extends to D9 and D10, where each model labels a consequence dimension on roughly six and eight of the eleven grounded instances the oracle reserves for a neighbor, depressing those active-convention dimensional means (F3, Appendix A) well below the recall the models actually achieve where the dimension is the truth. We flag the consequence-dimension boundaries for a sharpened v1.1 contract. On the empty-truth calibration instances, the residual false alarms are sparse redundancy and intent over-predictions by the two models evaluated on that split; no model invents sessions, prefixes, or blackholes out of nothing.

F3: The blind spot is the quiet half, not the second order. We expected models to read first-order table effects and miss second-order consequences. The data draw the line elsewhere. The *loud* half is read well across the fleet: five of the eight models—the four leaders plus the open-weight DeepSeek v4-pro—identify both grounded blackholes on every sample (10/10 each on *leak_ms* and *clos_summary*—the latter a genuinely second-order, partial-ECMP consequence of a summarization change), Sonnet 4.6 nearly matches them (D7 0.97), and prefix-loss (D2) F1 is 0.67–0.83 across that group; even the two cheapest models catch the explicit leak blackhole (Haiku 1.00 grounded, 0.98 on the templated leak-blackhole class), though they alone miss the implicit, topology-emergent one (*clos_summary*: Haiku 0.13, GPT-5.4-mini 0.13). Read where each *consequence* dimension is the ground truth, the picture is the same: the masked control-plane withdrawal (*ipc3_static*, D9) is caught at F1 1.00 by six models and 0.80/0.60 by the two cheapest; the management-plane lockout (*mgmt_lockout*, D10) at 1.00 by Opus, GPT-5.5, and Sonnet, 0.80 by both open-weight models (Haiku 0.76, Gemini 0.20); and the explicit LOCAL_PREF waypoint bypass (*lp_waypoint*, D4) at 1.00 by all eight (Figure 2).

The *quiet* half is the damage that survives this object-level reading: reachability holds, and the harm is a path or policy the diff never spells out. Its sharpest case is the leak-induced silent reroute, and it *separates the fleet* (Table 3). On the six reroute instances—where a more-specific leak

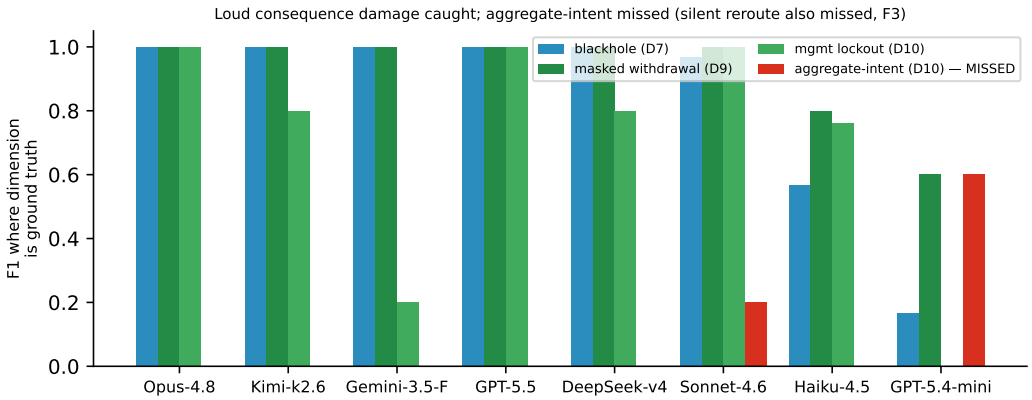


Figure 2: F1 where each consequence dimension is the ground truth (grounded, semantic grader; on these single-truth instances F1 tracks recall). The loud half—blackhole (D7), masked withdrawal (D9, *ipc3_static*), management lockout (D10, *mgmt_lockout*)—is caught by the stronger models; the aggregate-preservation intent (D10, *ipc_aggregate*) is missed. The sharpest quiet case, the leak-induced silent reroute, is missed (F3) and is not plotted here.

elsewhere pulls a victim's traffic past its mandated waypoint while the route still resolves and every destination stays reachable—most models score zero on *both* the waypoint and intent dimensions on the majority of samples: GPT-5.4-mini and Gemini 3.5 Flash on all 30, Haiku 4.5 on 27/30, Opus 4.8 on 21/30, and Sonnet 4.6 on 18/30—and Opus and Gemini do this *despite* topping the loud half (their D7/D9/D10 are at or near 1.00). Only two models register it at all: GPT-5.5 escapes total blindness on 29 of 30 samples (waypoint F1 0.64), and the open-weight DeepSeek v4-pro on 20 of 30 (0.56)—and even these reads are mostly of the waypoint *element*; the paired traffic-pinning *intent* key is missed far more often (credited on 19/30 of GPT-5.5's and 7/30 of DeepSeek's samples), so no model reliably reports the full reroute. Reading the silent reroute is nonetheless a distinct capability axis—uncorrelated with loud-half skill (Opus and Gemini top the loud half yet miss the reroute) and not vendor-bound: a proprietary model (GPT-5.5) and an open-weight one (DeepSeek) clear the field together (F4). This differentiation is grader-dependent: under naive exact-string matching, *every* model scores both-zero on all 30 (the same probe-granularity artifact as D7, §2); the semantic grader's canonicalization of the waypoint and intent keys is what exposes the split, so the headline rests on the matching specification for D4/D10—config-derived and floor-invariant (F5) but, unlike D1/D2/D8/D9, not provably the identity. A second, distinct quiet case is aggregate preservation: when a customer's specific prefixes are dropped but a covering /24 survives (*ipc_aggregate*), models catch the prefix loss (D2) yet miss the traffic-pinning intent the surviving aggregate masks (D10 F1 $\leq$0.20 for seven of the eight). The low *dimensional* means for the consequence dimensions (active-convention D7 0.17–0.40, D9 0.09–0.14, D10 0.02–0.11; Figure 1) are *not* recall failures: they are dominated by taxonomy-boundary false positives—each model predicts a consequence dimension on roughly three (D7), six (D9), and eight (D10) of the eleven grounded instances where the oracle reserves a neighboring label—the same over-prediction the paper discloses for the D3/D7 boundary (F2). The genuine miss is narrow and consistent: damage is read when it deletes or blackholes something nameable, and missed—by most of the fleet—when every object survives and only the path silently shifts. This is the path/policy-violation kernel of the more-specific-leak class (the 2019 Cloudflare/Verizon leak among them), which we model deterministically, isolated from the congestion and loss the real incidents also produced (§2). We measure the split on six seed-and-

Model	both-zero (total miss)	waypoint credited	intent credited	waypoint F1
GPT-5.5	1/30	29/30	19/30	0.64
DeepSeek v4-pro *(ow)*	10/30	20/30	7/30	0.56
Claude Sonnet 4.6	18/30	11/30	11/30	0.30
Claude Opus 4.8	21/30	8/30	4/30	0.26
Kimi k2.6 *(ow)*	*reroute run completing as of this draft*			
Claude Haiku 4.5	27/30	3/30	1/30	0.10
GPT-5.4-mini	30/30	0/30	0/30	0.00
Gemini 3.5 Flash	30/30	0/30	0/30	0.00

Table 3: The silent reroute separates the fleet (six instances, N=5, semantic grader). "both-zero" = samples scoring 0 on *both* the waypoint (D4) and intent (D10) dimensions (total blindness); "credited" = samples with nonzero F1 on that dimension. Two models clear the field (GPT-5.5 and the open-weight DeepSeek), statistically tied at the top (paired t=1.3, n=6 instances, n.s.) and well above the rest; the loud-half leaders Opus 4.8 and Gemini 3.5 Flash miss it. Even the leaders credit the traffic-pinning *intent* key far less often than the waypoint element, so no model reads the full reroute. Under exact-string matching every model scores both-zero on all 30 (granularity artifact, §2); the split is a semantic-grader result.

scope variants of a *single* leak-reroute mechanism; whether it generalizes across structurally distinct quiet-reroute mechanisms—LOCAL_PREF steering, AS-path manipulation, aggregate masking—is the central open question (§8).

F4: No home-field advantage. The benchmark was developed and red-teamed primarily with Claude-family models, so grader circularity was a live concern. Three lines of evidence rule it out. *(i) Frontier parity:* the cross-vendor leaders land within noise of one another—Opus 4.8 at 0.51, Kimi k2.6 at 0.49, Gemini 3.5 Flash at 0.49, GPT-5.5 at 0.48—a spread of 0.03 against a per-model standard error of ≈0.05 (n=11 instances), statistically indistinguishable (a Welch t-test on per-instance means does not reject equality, t=0.35). *(ii) No Claude tilt:* the second Anthropic model (Sonnet 4.6, 0.42) scores *below* four non-Anthropic models, and a non-Anthropic open-weight model (Kimi, 0.49) ties the best non-Anthropic closed model, so the Claude-built benchmark does not favor the Claude family. *(iii) On the headline:* the one capability the benchmark most sharply discriminates—reading the silent reroute (F3)—is cleared by two models, one proprietary (GPT-5.5) and one *open-weight* (DeepSeek v4-pro), statistically tied at the top (paired t=1.3, n=6 instances, n.s.) and both well above the rest. DeepSeek's waypoint F1 (0.56) exceeds every Anthropic and Google model's (Opus 0.26, Sonnet 0.30, Haiku 0.10, Gemini 0.00)—significantly over Opus (paired t=2.7, p=0.045) and marginally over the next-best Claude model Sonnet (t=2.2, p=0.08, n=6). The robust statement is that an open-weight reasoner stands among the two field-clearing models, so the quiet-half gap is neither a Claude-family nor a closed-model artifact—though on n=6 instances we do not claim a strict open-beats-all ordering.

F5: Naive set grading would have reported universal blindness. Under exact-string matching against the oracle's probe-granularity keys, every model's active-convention forwarding-anomaly F1 is 0.00 (no forwarding-anomaly-truth sample earns credit), and the grounded frontier macro drops from 0.48–0.51 to 0.36–0.37 (Appendix A)—while the raw outputs contain the correct node, kind, and config-visible prefix. The semantic canonicalization that corrects this is provably the identity on the config-visible dimensions (D1, D2, D8, D9) and leaves every model-free floor at

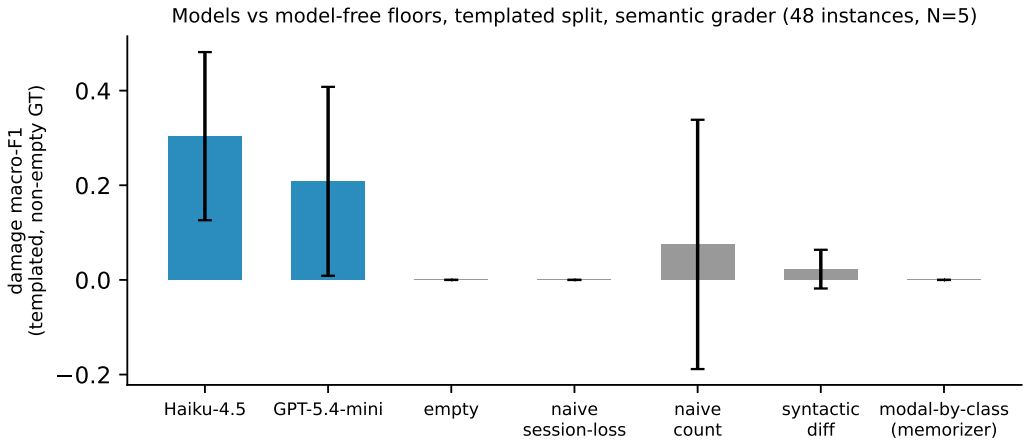


Figure 3: Damage-detection macro-F1 versus model-free floors on the templated split (semantic grader). The modal-by-class memorizer—the strongest template-recall strategy—scores 0.00.

chance (largest movement: the syntactic-diff floor, 0.007→0.023 templated), so the correction lifts models, not floors—and only on the dimensions where the raw outputs were right all along. We report this as a methods finding in its own right: granularity-mismatched string grading is a silent way for a verifier-graded benchmark to manufacture a dramatic, wrong headline.

6 Discussion

Why the quiet half matters most. An assistant that misread routing tables would be caught by any reviewer; an assistant that reads them competently, catches the withdrawal and the lockout, never false-alarms on sessions (F2), and yet silently misses the leak-induced reroute (F3)—as most of the evaluated fleet does—is trusted exactly where it is most dangerous—on the one failure mode that leaves no object lost to catch. Binary safety metrics cannot surface this: under a pass/fail rubric, a model that predicts "no impact" on a reroute instance is indistinguishable from one that flags the waypoint bypass, because both made *a* prediction. Typed, graded affected-set scoring is what makes the quiet half measurable—and the quiet half is where the most insidious incidents live: a leak that reroutes a victim's traffic onto a path it should never take, with every session up and every destination still reachable, exactly as in the 2019 route-leak outages.

A candidate graduation test for network agents. As an explicitly novel *position* (a refinement orthogonal to TM Forum's autonomous-network levels), we propose blast-radius prediction as a promotion gate on the autonomy ladder: *advise* → *propose* → *propose-with-self-check* → *gated-execute* → *autonomous*—a system would demonstrate, before advancing past *propose*, that it can enumerate the blast radius of its own changes at a calibrated precision/recall point *across all scored dimensions, including the quiet ones*. No evaluated model fully meets that bar today, but the data locate the frontier of the gate precisely. Every model passes on the loud half of these 4–6 node networks; on the quiet half the strongest reasoners are *beginning* to pass—GPT-5.5 reads the silent reroute on 29 of 30 samples (waypoint F1 0.64) and the open-weight DeepSeek v4-pro on 20 of 30 (0.56)—while most of the fleet, including the models that top the loud half (Opus 4.8, Gemini 3.5 Flash), still fails it outright. The gate therefore discriminates twice: loud from quiet, and—within

the quiet half—the reasoning frontier from the rest, which is the kind of resolution a promotion gate would need. We offer the autonomy ladder as a *position*, not a validated instrument: establishing blast-radius prediction as an actual gate requires the larger-topology and human-expert-baseline studies we defer (§8), and we conjecture—without testing it—that the quiet-half gap does not close as topologies grow, since the inference chains only lengthen.

Open problems. Five follow directly. (1) *Tools-augmented contrast*: how much of the quiet-half gap closes when the model may invoke the verifier, and do agents invoke it on the changes that need it? Cost-aware verifier invocation has been studied in general agentic workflows [13]; its networking instantiation—where a formal oracle defines the invocation-policy optimum—is open, and Asadli et al. point at analyzing agents' tool-usage patterns as future work [1]. (2) *Calibration*: are models *confidently* blind on the silent reroute, or do their confidences flag the gap? (3) *Judge validity*: would an LLM-as-judge grader have caught the granularity artifact of §2, or introduced its own? (4) *Learnability*: parameterized generators plus the oracle yield unlimited perfectly-labeled training pairs; whether supervision closes the quiet-half gap, or the gap survives infinite labels, is an open empirical question. (5) *Contract sharpening*: the D3/D7 boundary (F2) should be stated in the output contract precisely enough that double-typing is unambiguous, without leaking the oracle's disposition taxonomy.

7 Related Work

The rapidly growing use of LLMs across network management and operations is surveyed by Hong et al. [9]; that survey catalogs intent interpretation and task automation but does not treat pre-deployment safety validation—the gap NetBrace targets.

LLM configuration synthesis and repair. NetConfEval translates intents into specifications and code [6]; Mondal et al. study LLM router-configuration synthesis with verifier feedback [7]; PeeringLLM-Bench scores LLM synthesis of vendor-specific BGP peering configurations from policy intent [8]; CORNETTO benchmarks end-to-end configuration repair on 231 instances over real-world topologies of 20–754 nodes [2], and Asadli et al. show agentic architectures with retrieval and Batfish feedback improve repair efficacy ($\sim$12%) and reduce safety regressions ($\sim$17%) over monolithic prompting [1], where the safety metric is a *regression score*: the fraction of the checked specification set that an *applied* fix newly breaks, computed by the verifier (lower is better). NetBrace measures the complement: whether the *model itself*, with no verifier in the loop, can foresee typed damage from a diff. The verifier is referee, not teammate.

Agentic network benchmarks. NetArena (formerly NetPress) generates dynamic agent tasks and scores correctness, latency, and a rule-adherence action-safety axis in emulation [3]; NetAgentBench formalizes multi-turn configuration tasks as FSMs and measures behavioral stability [4]; NIKA evaluates incident diagnosis on emulated networks [5]; the IRTF NMRG framework draft defines NetConfBench, a 40-task emulator-based suite scored on intent-achievement and functional correctness [10]. Among these, none scores an LLM's *predicted* typed affected set—penalizing both missed and hallucinated damage at the element level—from a static diff, with no verifier in the model's loop. That conjunction is the gap NetBrace fills.

Evaluation methodology. Our design follows the rigor agenda for agentic benchmarks [14]: deterministic non-LLM grading, pinned oracles, contamination controls, explicit floors, and multi-

sample reporting. The granularity pitfall of §2 is offered as a concrete addition to that agenda. The holistic multi-metric framing descends from HELM [15].

8 Limitations

Single vendor dialect (FRR) and BGP-centric mechanisms in this release; the packet-filter dimension (D6) is deferred pending a vendor dialect whose ACLs Batfish models as interface filters. The D8 resource *budget* is a declared scenario parameter (the route count is computed; the limit is spec'd, as platform limits are). Topologies are small (4–6 nodes), chosen for oracle determinism; scale is untested, and while we see no mechanism by which the quiet-half reasoning gap would *close* as topologies grow, that expectation is a conjecture, not a result. The grounded split is 11 scenarios; several dimensions appear there once and are reported as point estimates, with the templated split carrying the statistics; the templated split currently has no redundancy-truth (D5) instance, so templated D5 reflects only false-positive behavior. Evaluation is single-shot and tool-free by design; agentic and tools-augmented conditions are the named next step. Batfish models converged steady state, so transient and timer-dependent damage is out of scope. Because grading is deterministic set comparison against a pinned verifier, there is no human-judgment step requiring inter-rater validation; the analogous validity question—does the oracle compute the right ground truth?—is addressed by the determinism gate, the live-FRR spot checks (§2), and the frozen matching specification. A human-*expert baseline* (can a senior operator enumerate the quiet half from the same evidence?) is the named next experiment and is not claimed here—it is the load-bearing control for reading the quiet-half miss as a *model* limitation rather than intrinsic task hardness. The headline silent-reroute result also rests on six seed-and-scope variants of a *single* leak-reroute mechanism; whether the loud/quiet split generalizes to structurally distinct quiet-reroute mechanisms (LOCAL_PREF steering, AS-path manipulation, aggregate masking) is the central open question and the first item a v1.1 must test. Relatedly, the reroute split is a *semantic-grader* result—under exact-string matching no model reads it (F3)—so the matching specification for the flow-scoped dimensions D4/D7/D10 is a config-derived, floor-invariant, model-independent design choice (F5) that, unlike the config-visible dimensions, has not been validated against independent human judgments; a grader-validity check on those canonicalization decisions is the natural complement to the human-expert baseline. The two open-weight models were evaluated on the grounded and silent-reroute classes but not on the cost-capped templated split, so their templated-floor comparison is deferred with the full-fleet templated run.

9 Conclusion

We introduced typed blast-radius prediction, built an oracle-graded, determinism-gated, judge-free benchmark for it, and measured where current LLMs sit. The line we found is not the one we expected: models—including mid-tier ones—catch the damage that announces itself, up to and including a second-order partial-ECMP blackhole, and never hallucinate a session loss; where most of them stumble—the one class that separates the reasoning frontier from the rest—is the damage that removes no object and only shifts the path: the leak-induced silent reroute past the inspection node, where the route still resolves, the session stays up, and reachability survives. Most of the fleet misses it even while catching the withdrawal, the lockout, and the blackhole, which all delete something nameable; only two models clear it—GPT-5.5 and the open-weight DeepSeek v4-pro, tied at the top and well above the rest—an open-weight model standing among the leaders, which rules out a closed-model or Claude-family artifact on the one capability the benchmark most

sharply discriminates. We also showed how easily a verifier-graded benchmark can manufacture the opposite, more dramatic headline through granularity-mismatched string grading, and we release both graders with a frozen matching specification so the correction is inspectable. Until the quiet half closes—or is closed by verifier-in-the-loop architectures whose invocation policies are themselves measured—LLM-proposed network changes need a blast-radius gate that the proposing model cannot currently pass on exactly the failures that matter most. The benchmark, oracle pipeline, run artifacts, and floors make that gate, and progress against it, measurable.

Artifacts. Code, scenario suites, generator seeds, pinned oracle digests, determinism reports, prompts, raw model outputs, graded results under both graders, and the frozen semantic-matching specification will be released alongside the peer-reviewed version; every number in this report traces to a saved run artifact (including the pooled Gemini sample set, both Sonnet runs, and the eight-model silent-reroute extension) and will be reproducible from that release.

Acknowledgments and AI disclosure. LLM tooling (Claude) was used extensively under the author's direction for drafting, engineering, and analysis, including adversarial review passes that surfaced the grading artifact reported in §2. All experiments, numbers, and citations were verified by the author, who takes sole responsibility for the content.

A Exact-string grading: the conservative lower bound

Table 4 reports the grounded split under both graders. The exact-string grader demands byte-identical element keys, including the oracle's undisclosed probe-/32 destinations on flow-scoped dimensions; the semantic grader (§2) canonicalizes probes to config-visible covers under a frozen, floor-invariant specification. Dimensions whose truth is already config-visible (D1, D2, D8, D9) are provably identical under both graders, as is the false-session-alarm metric.

Model	macro-F1 (damage) exact	macro-F1 (damage) semantic	forwarding D7 (active conv.) exact	forwarding D7 (active conv.) semantic
Claude Opus 4.8	0.37±0.27	0.51±0.17	0.00	0.40
Kimi k2.6 *(ow)*	0.36±0.24	0.49±0.12	0.00	0.40
Gemini 3.5 Flash	0.36±0.28	0.49±0.18	0.00	0.40
GPT-5.5	0.36±0.27	0.48±0.17	0.00	0.33
DeepSeek v4-pro *(ow)*	0.31±0.18	0.44±0.08	0.00	0.33
Claude Sonnet 4.6	0.30±0.17	0.42±0.06	0.00	0.32
Claude Haiku 4.5	0.29±0.17	0.39±0.10	0.00	0.23
GPT-5.4-mini	0.22±0.22	0.27±0.21	0.00	0.17

Table 4: Grounded split under both graders. The exact-string D7 column is 0.00 for every model on every forwarding-anomaly-truth sample—the granularity artifact of §2. The semantic D7 column uses the active-instance convention and is depressed below the anomaly-truth recall of Table 2 by the D3/D7 double-typing false positives discussed in F2. Sonnet's grounded rows are from the 12,000-token re-run (Disclosure ii); the exact-string column moves it from 0.30 to 0.42 under semantic matching, the same lift seen fleet-wide.

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